

Alex Khundongbam

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PROFILE

AI Engineer specializing in autonomous agent systems, RAG pipelines, and production LLM applications. Experienced in designing and deploying multi-agent architectures using LangChain, LangGraph, and CrewAI, with hands-on expertise in async Python, FastAPI microservices, and workflow orchestration. Skilled in hallucination mitigation, agent evaluation frameworks, and translating complex AI capabilities into reliable, enterprise-grade automation systems.

WORK EXPERIENCE

AI and Data Engineer

Jan 2026 – Present

LeadWalnut

Bangalore

- Designed and deployed autonomous document processing agents using event-driven triggers and API-driven workflow orchestration (n8n), handling 100+ invoice documents/day with end-to-end validation and error recovery.
- Architected multi-stage AI pipelines with structured output parsing, JSON extraction, and schema validation layers — reducing hallucinations and extraction errors by enforcing strict output contracts without relying on ungarded LLM responses.
- Built async FastAPI microservices for scalable document ingestion, agent task routing, and accounting system integration, ensuring non-blocking throughput under concurrent document loads.
- Implemented agent monitoring and observability systems including structured logging, workflow state tracking, and failure alerting to ensure production reliability of automation agents.
- Developed Python-based AI and analytics tools analyzing 50k+ SEO and CRM records to support keyword intelligence, competitor analysis, and marketing performance optimization.
- Defined and enforced Non-Functional Requirements (NFRs) — data validation, error handling, logging, and retry logic — to ensure production readiness of all automation workflows.

AI Engineering Intern – ERP Systems

Jul 2025 – Dec 2025

Christ University, Centre for Digital Innovation, Bangalore

- Designed and deployed async FastAPI-based AI microservices for document processing, form automation, and internal reporting workflows integrated with the university ERP system.
- Implemented agent safety guardrails and output validation layers to ensure reliable and consistent responses from LLM-powered features in production.
- Improved system observability through structured logging and monitoring dashboards, enabling rapid identification of agent failures and performance bottlenecks.

AI/ML R&D Intern

Nov 2024 – May 2025

Christ University, Bangalore

- Developed and orchestrated autonomous AI agents using LangChain, LangGraph, and n8n — integrating tool use, vector store retrieval, and document loaders to build functional multi-agent pipelines.
- Built and evaluated RAG systems using Pinecone and Chroma vector databases, applying chunking strategies and embedding optimization to improve retrieval precision and reduce irrelevant context injection.
- Applied Chain-of-Thought (CoT) prompting and PEFT fine-tuning for performance optimization of transformer models, including hallucination reduction through structured prompt constraints.

Data Science Intern

Nov 2023 – Feb 2024

Solar Secure Solutions

Remote, Bengaluru, India

- Processed and cleaned 500k+ data points to improve data quality and enhance downstream model performance and reliability.
- Developed a churn prediction model achieving 95% accuracy, enabling proactive customer retention strategies.
- Built a recommendation system that increased user engagement by 20% through personalized content and behavioral analysis.
- Designed a time-series forecasting model to support demand planning and improve operational decision-making.

SKILLS

LLM & Agentic AI: LangChain, LangGraph, CrewAI, AutoGen, RAG, PEFT, Prompt Engineering, Hallucination Mitigation, CoT Prompting
Agent Evaluation & Safety: Output Validation, Guardrails, RAGAS Evaluation, Structured Output Contracts, Agent Monitoring
MLOps & Deployment: FastAPI, Flask, Docker, Weights & Biases, AWS, GCP, Vercel, CI/CD Pipelines
Data & Analytics: Pandas, NumPy, Spark, Scikit-Learn, PyTorch, TensorFlow, XGBoost
Vector Databases: Pinecone, Chroma, Weaviate
Languages & Systems: Python (asyncio), SQL, JavaScript, Java, C, Microservices, API Design, Event-driven Architecture

PROJECTS

EverMind LongContext Agent: Built a conversational agent with hybrid memory (SQLite + ChromaDB) that compresses long context 3:1 while keeping 96% Q&A accuracy and sub-1.5s response latency, using async/parallel processing and OpenAI embeddings.

SpeakFiction: Built a RAG-based conversational AI system enabling users to interact with fictional characters from uploaded books. Implemented chunking, embedding optimization, and context injection strategies to minimize hallucinations and improve retrieval faithfulness.

Christ IntelliTeach: Built and deployed a data-driven web platform for educators with automated LLM-powered content generation, structured document processing, and usage analytics. Integrated async API services for concurrent multi-user support.

CompressLM — LLM Compression & Distillation Pipeline: Compressed a 3B-parameter instruct LLM into a 0.5B student via LoRA + KL-divergence distillation, then INT4-quantized it — 19x smaller and 7.5x faster inference, retaining 93% teacher-level accuracy and 97% JSON-schema validity.

RESEARCH

Systematic Evaluation of Vision Transformer Robustness under Gradient, Optimization, and Black-Box Attacks with Hybrid Defense Mechanisms

Classic Models, Modern Threats: A Study on Adversarial Attack and Defense for Traditional ML Models

Investigating the Impact of Feedback Loop Training on Machine Learning Model Performance in Time Series Data

Assessing the Impact of COVID-19 on Inflation Dynamics in India: A Machine Learning Approach

EDUCATION

Christ University, Central Campus	Bangalore
<i>M.Sc. in Artificial Intelligence and Machine Learning</i>	2024 – Present
Kristu Jayanti College	Bangalore
<i>B.Sc. in Physics and Computer Science – 8.45 CGPA — First Rank Holder</i>	2021 – 2024